



sensores ▾

bomba_de_agua / bomba_agua / sensores.ino



luizasilva08 Atualização do código dos sensores

Code

Blame

58 lines (49 loc) · 2.11 KB



```
1  void setupSensores() {
2      pinMode(PINO_S1, INPUT_PULLUP);
3      pinMode(PINO_S2, INPUT_PULLUP);
4      pinMode(PINO_S3, INPUT_PULLUP);
5      pinMode(PINO_S4, INPUT_PULLUP);
6      pinMode(PINO_S5, INPUT_PULLUP);
7  }
8
9  void lerSensores(unsigned long tempoAtual) {
10     bool s1_raw = (digitalRead(PINO_S1) == LOW);
11     bool s2_raw = (digitalRead(PINO_S2) == LOW);
12     bool s3_raw = (digitalRead(PINO_S3) == LOW);
13     bool s4_raw = (digitalRead(PINO_S4) == LOW);
14     bool s5_raw = (digitalRead(PINO_S5) == LOW);
15
16     bool s1_valido = s1_raw;
17     bool s2_valido = s1_raw && s2_raw;
18     bool s3_valido = s1_raw && s2_raw && s3_raw;
19     bool s4_valido = s1_raw && s2_raw && s3_raw && s4_raw;
20     bool s5_valido = s1_raw && s2_raw && s3_raw && s4_raw && s5_raw;
21
22     erroSensores = false;
23     sensorComErro = 0;
24
25     if (s2_raw && !s1_raw) { erroSensores = true; sensorComErro = 1; }
26     else if (s3_raw && !s2_valido) { erroSensores = true; sensorComErro = 2; }
27     else if (s4_raw && !s3_valido) { erroSensores = true; sensorComErro = 3; }
28     else if (s5_raw && !s4_valido) { erroSensores = true; sensorComErro = 4; }
29
30     if (!erroSensores) {
31         float limiteAcionado = 0.0;
32         if (s5_valido) { nivel = 5; porcentagem = 85; limiteAcionado = 8500.0; }
33         else if (s4_valido) { nivel = 4; porcentagem = 60; limiteAcionado = 6000.0; }
34         else if (s3_valido) { nivel = 3; porcentagem = 45; limiteAcionado = 4500.0; }
35         else if (s2_valido) { nivel = 2; porcentagem = 30; limiteAcionado = 3000.0; }
36         else if (s1_valido) { nivel = 1; porcentagem = 15; limiteAcionado = 1500.0; }
37         else { nivel = 0; porcentagem = 0; limiteAcionado = 0.0; }
38     }
```



bomba ▾

bomba_de_agua / bomba_agua / bomba.ino



anacrabreu-dot código da bomba de agua (servo motor)

Code

Blame

41 lines (35 loc) · 1.33 KB



```
1  void setupMotorEEPROM() {
2      servoBomba.attach(PINO_SERVO);
3
4      EEPROM.get(enderecoConsumoTotal, consumoTotal);
5      if (isnan(consumoTotal) || consumoTotal < 0.0 || consumoTotal > 99999.0) consumoTotal = 0.0;
6
7      EEPROM.get(enderecoVolumeAtual, volumeAtual);
8      if (isnan(volumeAtual) || volumeAtual < 0.0) volumeAtual = 0.0;
9      if (volumeAtual > 8500.0) volumeAtual = 8500.0;
10
11     EEPROM.get(enderecoPorcentagem, porcentagem);
12     if (porcentagem < 0 || porcentagem > 100) porcentagem = 0;
13
14     if (porcentagem >= 85) nivel = 5;
15     else if (porcentagem >= 60) nivel = 4;
16     else if (porcentagem >= 45) nivel = 3;
17     else if (porcentagem >= 30) nivel = 2;
18     else if (porcentagem >= 15) nivel = 1;
19     else nivel = 0;
20 }
21
22 void controlarBomba() {
23     // s5_valido não existe diretamente aqui, mas sabemos que se nível == 5, a caixa tá cheia.
24     // Vamos usar a lógica de porcentagem para ser mais seguro, ou checar o erro.
25     if (nivel == 5 || erroSensores) {
26         bombaLigada = false;
27         servoBomba.write(90);
28     } else {
29         bombaLigada = true;
30         servoBomba.write(180);
31     }
32 }
33
34 void salvarDados(unsigned long tempoAtual) {
35     if (tempoAtual - tempoAnteriorSalvar >= 5000) {
36         EEPROM.put(enderecoConsumoTotal, consumoTotal);
37         EEPROM.put(enderecoVolumeAtual, volumeAtual);
38         EEPROM.put(enderecoPorcentagem, porcentagem);
39         tempoAnteriorSalvar = tempoAtual;
40     }
41 }
```

display bomba_de_agua / bomba_agua / display.ino

matheusjardim-a11y Display

Code Blame 44 lines (40 loc) · 1.2 KB

```
1 void setupDisplay() {
2     lcd.init();
3     lcd.backlight();
4     lcd.setCursor(0, 0);
5     lcd.print("Inicializando...");
6     lcd.setCursor(0, 1);
7     lcd.print("Aguarde 5 seg.");
8     delay(5000);
9     lcd.clear();
10 }
11
12 void atualizarDisplay(unsigned long tempoAtual) {
13     if (tempoAtual - tempoAnteriorLCD >= 300) {
14         tempoAnteriorLCD = tempoAtual;
15         lcd.setCursor(0, 0);
16
17         if (erroSensores) {
18             lcd.print("ERRO NO SENSOR "); lcd.print(sensorComErro);
19         } else {
20             lcd.print("N:"); lcd.print(nivel); lcd.print(" ");
21             if (porcentagem < 10) lcd.print("0");
22             lcd.print(porcentagem); lcd.print("% V:");
23             if (volumeAtual < 10) lcd.print("000");
24             else if (volumeAtual < 100) lcd.print("00");
25             else if (volumeAtual < 1000) lcd.print("0");
26             lcd.print((int)volumeAtual); lcd.print("L ");
27         }
28
29         lcd.setCursor(0, 1);
30         lcd.print("Cons:");
31         unsigned long cons = (unsigned long)consumoTotal;
32         if (cons > 9999) lcd.print("+9k ");
33         else {
```

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).

base: main ← compare: sensores ✓ **Able to merge.** These branches can be automatically merged.

Discuss and review the changes in this comparison with others. [Learn about pull requests](#)

Create pull request

→ 4 commits

2 files changed

By 1 contributor

Commits on Mar 19, 2026

Inserindo código dos sensores

luizsilva08 committed 5 days ago

7448389 <>

código para aparecer no Serial Monitor

luizsilva08 authored 5 days ago

Verified e9649b4 <>

Commits on Mar 24, 2026

Atualização do código conforme a main

luizsilva08 authored 2 hours ago

Verified 6ef1de0 <>

Atualização do código dos sensores

luizsilva08 authored 2 hours ago

Verified d6edf46 <>

Showing 2 changed files with 80 additions and 22 deletions.

Split Unified

```
44 bomba_agua/bomba_agua.ino
@@ -2,43 +3,43 @@
3 #include <EEPROM.h>
4 #include <Servo.h>
5
6 - // --- VARIÁVEIS GLOBAIS ---
7 LiquidCrystal_I2C lcd(Rx27, 16, 2);
8 Servo servoBomba;
9
10 const int PINO_POT_NIVEL = A0;
11 const int PINO_SERVO = 0;
12 - const int PINO_S1 = 6;
13 - const int PINO_S2 = 7;
14 - const int PINO_S3 = 9;
15 - const int PINO_S4 = 10;
16
17 #include <EEPROM.h>
18 #include <Servo.h>
19
20 + // --- OBJETOS E CONSTANTES GLOBAIS ---
21 LiquidCrystal_I2C lcd(Rx27, 16, 2);
22 Servo servoBomba;
23
24 const int PINO_POT_NIVEL = A0;
25 const int PINO_SERVO = 0;
26 + const int PINO_S1 = 6, PINO_S2 = 7, PINO_S3 = 9, PINO_S4 = 10, PINO_S5 = 11;
27 + const int enderecoConsumoTotal = 0, enderecoVolumeAtual = 4, enderecoPorcentagem = 5;
```

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).

base: main ← compare: bomba ✓ **Able to merge.** These branches can be automatically merged.

Discuss and review the changes in this comparison with others. [Learn about pull requests](#)

Create pull request

→ 1 commit

1 file changed

By 1 contributor

Commits on Mar 24, 2026

código da bomba de água (servo motor)

anacabreu-dot committed 2 hours ago

52462db <>

Showing 1 changed file with 41 additions and 0 deletions.

Split Unified

```
41 bomba_agua/bomba.ino
@@ -0,0 +1,41 @@
...
1 + void setupMotorEEPROM() {
2 +   servoBomba.attach(PINO_SERVO);
3 +
4 +   EEPROM.get(enderecoConsumoTotal, consumoTotal);
5 +   if (isnan(consumoTotal) || consumoTotal < 0.0 || consumoTotal > 9999.0) consumoTotal = 0.0;
6 +
7 +   EEPROM.get(enderecoVolumeAtual, volumeAtual);
8 +   if (isnan(volumeAtual) || volumeAtual < 0.0) volumeAtual = 0.0;
9 +   if (volumeAtual > 8500.0) volumeAtual = 8500.0;
10 +
11 +   EEPROM.get(enderecoPorcentagem, porcentagem);
12 +   if (porcentagem < 0 || porcentagem > 100) porcentagem = 0;
13 +
14 +   if (porcentagem >= 85) nivel = 5;
15 +   else if (porcentagem >= 60) nivel = 4;
16 +   else if (porcentagem >= 45) nivel = 3;
17 +   else if (porcentagem >= 30) nivel = 2;
18 +   else if (porcentagem >= 15) nivel = 1;
19 +   else nivel = 0;
20 + }
21 +
22 + void controlarBomba() {
23 +   // Se não existe diretamente aqui, mas sabemos que se nível == 5, a caixa tá cheia.
24 +   // Vamos usar a lógica de porcentagem para ser mais seguro, ou checar o erro.
25 +   if (nivel == 5 || erroSensores) {
26 +     bombaligada = false;
27 +     servoBomba.write(90);
28 +   } else {
29 +     bombaligada = true;
30 +     servoBomba.write(180);
31 +   }
32 + }
```

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).

base: main ← compare: display ✓ Able to merge. These branches can be automatically merged.

Discuss and review the changes in this comparison with others. [Learn about pull requests](#)

Create pull request

↔ 1 commit

📄 1 file changed

👤 1 contributor

Commits on Mar 24, 2026

Display

matheusjardim-atly committed 7 minutes ago

📄

esec22

<>

Showing 1 changed file with 44 additions and 0 deletions.

Split Unified

44 bomba_agua/display.ino

@@ -0,0 +1,44 @@

```
1 + void setupDisplay() {
2 +   lcd.init();
3 +   lcd.backlight();
4 +   lcd.setCursor(0, 0);
5 +   lcd.print("Inicializando...");
6 +   lcd.setCursor(0, 1);
7 +   lcd.print("Aguarde 5 seg.");
8 +   delay(5000);
9 +   lcd.clear();
10 + }
11 +
12 + void atualizarDisplay(unsigned long tempoAtual) {
13 +   if (tempoAtual - tempoAnteriorCD >= 300) {
14 +     tempoAnteriorCD = tempoAtual;
15 +     lcd.setCursor(0, 0);
16 +
17 +     if (erroSensores) {
18 +       lcd.print("ERRO NO SENSOR "); lcd.print(sensorComErro);
19 +     } else {
20 +       lcd.print("M:"); lcd.print(nivel); lcd.print(" ");
21 +       if (porcentagem < 10) lcd.print("0");
22 +       lcd.print(porcentagem); lcd.print("% V:");
23 +       if (volumeAtual < 10) lcd.print("000");
24 +       else if (volumeAtual < 100) lcd.print("00");
25 +       else if (volumeAtual < 1000) lcd.print("0");
26 +       lcd.print((int)volumeAtual); lcd.print("L ");
27 +     }
28 +
29 +     lcd.setCursor(0, 1);
30 +     lcd.print("Cons:");
31 +     unsigned long cons = (unsigned long)consumoTotal;
32 +     if (cons > 9999) lcd.print("+9k ");
```

Merge da branch sensores com a main #1

✓ Ready to merge

Code

Open luizasilva08 wants to merge 4 commits into main from sensores

Conversation 0

Commits 4

Checks 0

Files changed 2

+80 -22



luizasilva08 commented now

Collaborator

Fazendo o merge das duas branches para unir o código.



luizasilva08 added 4 commits 5 days ago



inserindo código dos sensores

7440389



código para aparecer no Serial Monitor

Verified

e8649b4



Atualização do código conforme a main

Verified

6ef1de0



Atualização do código dos sensores

Verified

d6edf46



✓ No conflicts with base branch

Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions](#).

Still in progress? [Convert to draft](#)

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

Customize

Subscribe

You're not receiving notifications from this thread.



Add a comment

codigo da bomba de agua (servo motor) #2

Open

luizasilva08 wants to merge 1 commit into main from bomba

Conversation 0

Commits 1

Checks 0

Files changed 1

+41

luizasilva08 commented now

Collaborator

No description provided.

codigo da bomba de agua (servo motor)

52462db

No conflicts with base branch

Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

codigo da bomba de agua (servo motor) #2

Merged

luizasilva08 merged 1 commit into main from bomba now

Conversation 0

Commits 1

Checks 0

Files changed 1

luizasilva08 commented now

Collaborator

No description provided.

codigo da bomba de agua (servo motor)

52462db

luizasilva08 merged commit f6a731d into main now

Revert

Pull request successfully merged and closed

You're all set — the `bomba` branch can be safely deleted.

Delete branch

código do Display #3

Open

luizasilva08 wants to merge 1 commit into `main` from `display`

Conversation 0

Commits 1

Checks 0

Files changed 1



luizasilva08 commented now

Collaborator ...

No description provided.



Display

e5ecc22



No conflicts with base branch

Merging can be performed automatically.

Merge pull request



You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)

código do Display #3

Open

luizasilva08 wants to merge 1 commit into `main` from `display`

Conversation 0

Commits 1

Checks 0

Files changed 1



luizasilva08 commented now

Collaborator ...

No description provided.



Display

e5ecc22



Pull request successfully merged and closed

You're all set — the `display` branch can be safely deleted.

Delete branch



Add a comment